

Perception of Cognitive Robots Milestone Report I

Nutdanai (6538049921), Wongsakorn (6538177521), Withawin (6538191221)

1 Introduction / Overview

This project investigates robotic perception through the development of a simulated mobile robot capable of understanding and interacting with its environment. The final system will integrate perception, localization, mapping, and navigation to enable autonomous goal-seeking behavior while avoiding obstacles.

This report presents **Milestone 1**, which focuses on **computer vision**. A monocular camera is used to distinguish static environmental structures from dynamic objects based on visual motion cues. All computer vision code for this milestone is available at:

<https://github.com/pepepepepordu/perception-of-cognitive-robot-project>

2 Robot Simulation

The system is implemented in the Webots simulation framework. The environment consists of a bounded arena with static walls and moving objects. An e-puck differential-drive robot equipped with a monocular camera serves as the sensing platform. During this milestone, the robot remains stationary to eliminate ego-motion and isolate visual motion generated by dynamic entities in the environment.

3 Computer Vision

3.1 Mathematical Model

The camera image is modeled as a luminance function $I(x, y, t)$. Motion estimation assumes brightness constancy:

$$I(x, y, t) = I(x + u, y + v, t + 1)$$

Applying a first-order Taylor expansion yields the optical flow constraint:

$$I_x u + I_y v + I_t = 0$$

Motion is estimated by minimizing the mean squared error over a discrete set of candidate displacements:

$$(u, v) \in \{(-1, 0), (1, 0), (0, -1), (0, 1), (0, 0)\}$$

$$E(u, v) = \frac{1}{N} \sum (I_x u + I_y v + I_t)^2$$

3.2 Method of Implementing

Each frame is converted to grayscale and smoothed using a Gaussian filter. Salient feature points are selected using the **Shi-Tomasi Corner Detector**, implemented via the *Good Features to Track* method. Local 5×5 neighborhoods are used to compute spatial and temporal gradients. Feature points with insufficient gradient magnitude are discarded. Points with non-zero estimated motion are classified as dynamic, while those converging to $(0, 0)$ are classified as static.

4 Implementation and Demonstration

The perception pipeline operates in real time, processing consecutive frames, extracting feature points, estimating motion, and classifying environmental entities. Console outputs confirm consistent detection of dynamic objects and correct identification of static structures using visual information alone.

5 Future Work

The next milestone, scheduled for **6 March**, focuses on **Simultaneous Localization and Mapping (SLAM)**. The robot will construct a map of the environment while estimating its own position.